PAPERGAMES

Nim: who takes the last stone?

Finding a strategy

Open the game

Name: _____

Goal: You learn to think ahead and find a safe winning strategy.

Rules: There are heaps of stones. On your turn you take as many stones as you like from **one** heap, at least one. Whoever takes the last stone wins.

1. One heap with 5 stones. It is your turn. How many stones do you take to win at once?

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2. Two heaps with 2 stones each. It is your turn. Can you be sure to win if the other player makes no mistake? Explain.

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3. Two heaps with 4 and 2 stones. It is your turn. Which move makes sure you win?

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4. Three heaps with 3, 4 and 5 stones (this is how Nim starts in PaperGames). Find the one move that makes sure you win. Hint: try to leave a position where every move of the other player helps you.

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Free from PaperGames. You may copy it freely for lessons. https://yourchaoschris.github.io/PaperGames/

PAPERGAMES

Solutions for teachers

Nim: who takes the last stone?

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1. Take all 5 stones.
2. No. Whatever you take, the other player can take the same from the other heap (mirroring). Then they take the last stone.
3. Take 2 stones from the heap of 4. Now there are 2 and 2, and from now on you mirror every move.
4. Take 2 stones from the heap of 3. That leaves 1, 4 and 5. For older students: write the heaps in binary (1 = 001, 4 = 100, 5 = 101). Every column then has an even number of ones. From such a position the player to move loses against correct play (Bouton, 1901).

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